📊 *Sales Data Analysis and Visualization Report*

1. Introduction

This report presents an analysis of the sales dataset (*awesome.csv*). The objective is to understand sales patterns, identify key factors affecting revenue, and derive actionable insights using statistical and visual analysis techniques.

2. Objective

The main objectives of this analysis are:

- To analyze sales (Amount) distribution
- To evaluate performance across different geographies
- To identify top-performing salespersons
- To study the relationship between Units and Amount
- To apply statistical methods such as hypothesis testing and confidence interval

3. Dataset Description

The dataset contains the following columns:

- **Sales Person** – Name of salesperson
- **Geography** – Region of sales
- **Product** – Product sold
- **Amount** – Revenue generated
- **Units** – Number of units sold

4. Data Preprocessing

- Checked for missing values (none/minimal)
- Verified data types (Amount and Units are numerical)
- Removed/handled any inconsistencies if present

5. Exploratory Data Analysis (EDA)

5.1 Univariate Analysis

- The distribution of **Amount** is **right-skewed**
- Most transactions are small to medium in value
- Few large transactions act as **outliers**

✓ Insight:
Revenue is influenced by a small number of high-value transactions

5.2 Geography Analysis

- Sales are distributed across multiple regions
- Median values are similar across regions

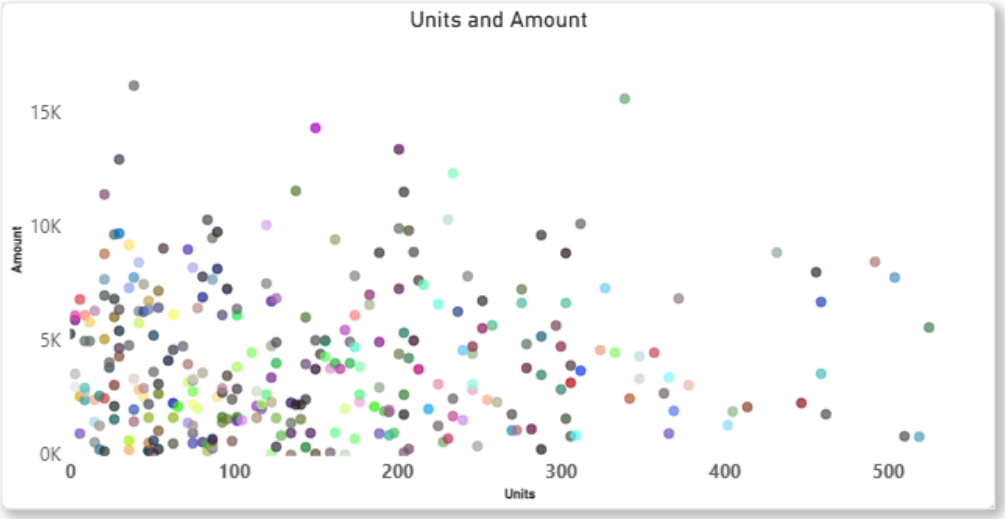
✓ Insight:
No major variation in sales performance across geographies

5.3 Sales Person Analysis

- A few salespersons contribute significantly higher revenue
- Others have comparatively lower performance

✓ Insight:
Sales performance is uneven, indicating scope for improvement

5.4 Bivariate Analysis (Units vs Amount)



- Scatterplot shows a **positive relationship**

✓ Insight:
Higher units sold lead to higher revenue

5.5 Correlation Analysis

- Positive correlation between **Units and Amount**

✓ Insight:
Units is a key factor influencing revenue

5.6 Outlier Detection

- Boxplot reveals presence of high-value outliers

✓ Insight:
Outliers may represent bulk orders or premium customers

6. Statistical Analysis

6.1 Hypothesis Testing

Test: USA vs UK Sales

- **H_0:** No difference in average Amount
- **H_1:** There is a difference

Result:

- Confidence Interval includes 0

✓ Conclusion:
There is **no statistically significant difference** between USA and UK sales

6.2 Confidence Interval

- Mean Amount ≈ **4136**
- 95% CI: **(3781, 4491)**

✓ Interpretation:
We are 95% confident that the true average sales amount lies within this range

7. Key Insights

- Sales are positively influenced by the number of units sold
- Revenue is concentrated among a few high-value transactions
- Sales performance varies among salespersons
- No significant geographical difference in sales
- Outliers play an important role in total revenue

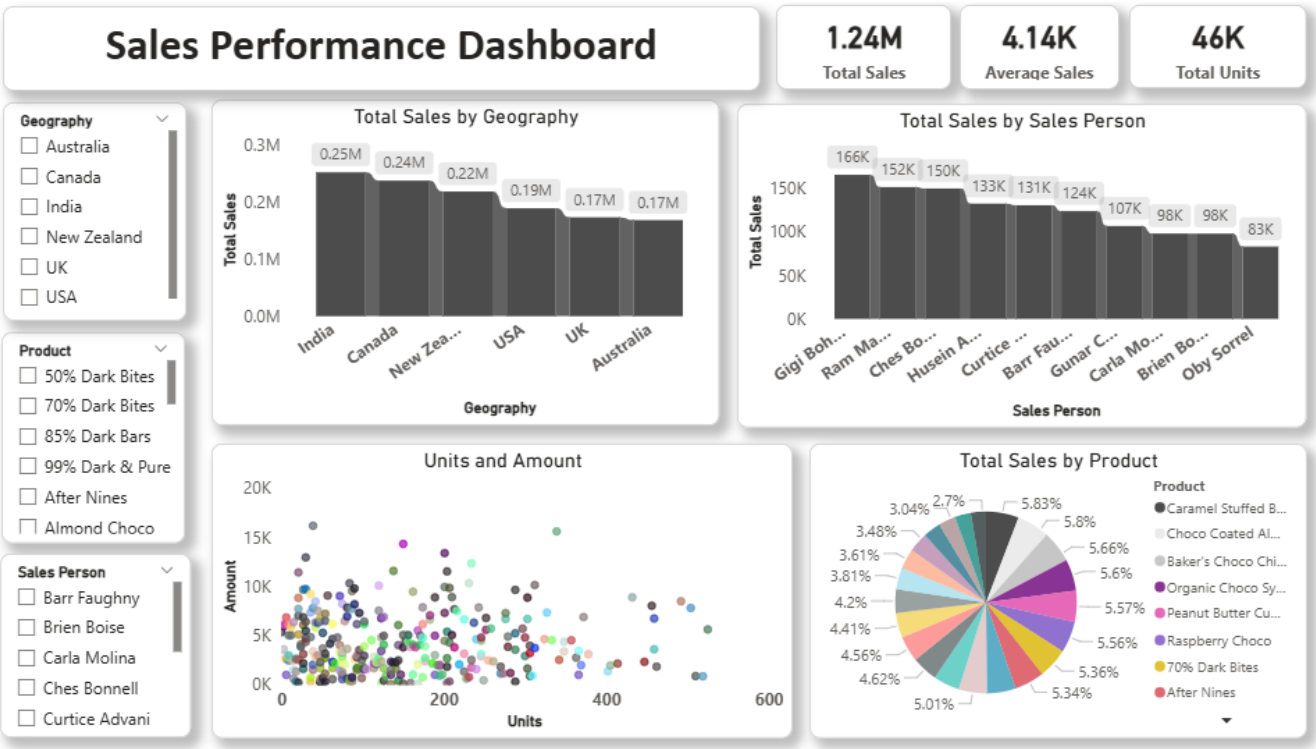
8. Dashboard Section

An interactive dashboard was developed using Microsoft Power BI to visualize sales performance and key insights from the dataset.

Key Features of Dashboard:

- KPI cards showing Total Sales, Total Units, and Average Sales
- Bar charts representing sales by Geography and Sales Person
- Scatter plot showing relationship between Units and Amount
- Pie chart showing product-wise sales distribution
- Slicers for dynamic filtering by Geography, Product, and Sales Person

Purpose:
The dashboard provides a clear and interactive view of sales data, enabling quick decision-making and performance evaluation.



9. Conclusion

The analysis shows that while sales performance is consistent across regions, it is highly influenced by units sold and a few high-value transactions. Salesperson performance varies, indicating opportunities for training and optimization. Statistical analysis confirms that geographical differences are not significant.

10. Recommendations

- Focus on increasing units sold to boost revenue
- Identify and retain high-value customers
- Train underperforming salespersons
- Analyze outliers separately for business strategy
- Expand successful sales strategies across all regions